

Solving & Graphing Inequalities

Use the cue, then fade the cue. Look at the cue card for 60 seconds. Then cover it before answering Round 1. Check the card only after you commit to an answer.

CUE FOCUS Solve an inequality like an equation, but FLIP the sign when you multiply or divide by a negative: $-2x < 6 \rightarrow x > -3$. To graph: open circle for $<$ or $>$, closed circle for $\leq$ or $\geq$, then shade the solution ray.

Round 1 — Retrieve It

1. When do you flip the inequality sign?

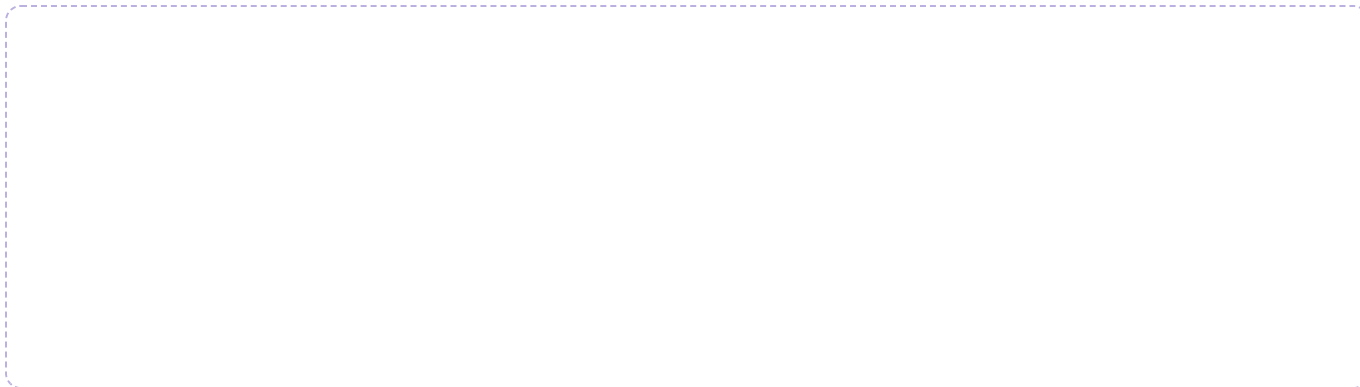
2. Solve: $-2x < 6$.

3. When do you use an open circle, and when do you use a closed circle?

4. After you place the circle, what do you do to finish the graph?

Round 2 — Sketch It

Draw a number line and graph $x > -3$. Use the correct circle for $>$ and shade the ray in the correct direction. Label the point -3 .



Solving & Graphing Inequalities — Apply It

Keep the cue card covered. Solve first. Then use the cue to check, revise, and explain what you changed.

Round 3 — Apply the Relationship

1. Solve and state the circle type: $x + 5 \leq 9$.

2. Solve: $3x > 12$.

3. Solve, and say whether the sign flips: $-4x \geq 20$.

Round 4 — Reason from Evidence

A student solved $-3x < 9$ and got $x < -3$. Find the error, give the correct solution, and explain the rule that was missed. Then describe the graph.

Solving & Graphing Inequalities — ACE It

Use the cue card only if you need it. The goal is to show what you can explain, connect, and use on your own.

A**Articulate**

Explain why the inequality sign flips when you divide by a negative number. Use one example that is not printed on the cue card.

C**Connect**

Connect an inequality to a real limit, such as "you must be at least 48 inches to ride." Write the inequality and describe its graph.

E**Extend**

Create your own inequality that requires flipping the sign. Solve it, then sketch the correct graph with the right circle and shading.

Confidence check. Circle one after you finish: I still need the cue / I need it once in a while / I can explain this without it.

Teacher Key — Solving & Graphing Inequalities

Remove this page before student copying. Answers below give expected ideas, not required wording.

Retrieve It

1. When you multiply or divide both sides by a negative number.
2. $-2x < 6 \rightarrow$ divide by -2 and flip: $x > -3$.
3. Open circle for $<$ or $>$ (not included); closed circle for $\leq$ or $\geq$ (included).
4. Shade the ray toward all values that make the inequality true.

Sketch It — look for

Open circle at -3 , shading extending to the right (toward larger numbers) for $x > -3$.

Apply It

1. $x + 5 \leq 9 \rightarrow x \leq 4$, closed circle.
2. $3x > 12 \rightarrow x > 4$ (no flip; divided by positive 3).
3. $-4x \geq 20 \rightarrow$ divide by -4 and flip: $x \leq -5$ (sign flips).

Reason from Evidence — look for

Dividing by -3 must flip the sign. $-3x < 9 \rightarrow x > -3$. Graph: open circle at -3 , shaded to the right.

ACE — Extend look-for

Many answers possible. Check that the inequality truly needs a flip, the solution is correct, and the graph shows the right circle and shading direction.

Suggested use. Run Page 1 with the cue card available, Page 2 with the cue mostly covered, and Page 3 as independent reflection. Let students uncover the cue to self-correct in a different color or with a revision mark.